Homework 2: Qualitative Analysis I — The Workflow

Paper track. Pairs with Home Lab 2. Do this on your own.

Part A: Identify from results (2 pts each)

For each unknown, give the **name and formula**.

#	Look	Water	pH	+ Acid	+ Iodine	Heat	Identity + formula
1	fine white powder	stays cloudy	~9	strong fizz	no change	just sits	
2	cubic crystals	dissolves fully	7	no fizz	no change	crackles, no melt	
3	fine white powder	"oobleck"	7	no fizz	**blue-black**	chars	
4	white powder	dissolves fully	~9	strong fizz	no change	just sits	
5	glassy grains	dissolves fully	7	no fizz	no change	melts, caramel smell	

Part B: Read the key (5 pts)

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1. Fizzes strongly with acid?      yes → 2
no → 3
2. Dissolves fully in water?        yes → BAKING
SODA        no → CHALK
3. Blue-black with iodine?          yes →
CORNSTARCH         no → 4
4. Dissolves fully in water?        no → GYPSUM
yes → 5
5. Cubic crystals, no melt on heat? yes → TABLE
SALT           no → SUGAR
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Which numbered steps do you pass through, and what's the answer, for:

1. A powder that fizzes with acid and leaves cloudy undissolved residue.
2. A powder that doesn't fizz, is iodine-negative, dissolves fully, and melts to a brown syrup on heat.
3. A powder that doesn't fizz and turns iodine blue-black.

Part C: Build a key (4 pts)

You have four unknowns and know the list is: **Epsom salt, citric acid, corn starch, table salt**. Write a dichotomous key (first question, branches, down to all four). *Hint: which one test cleanly removes corn starch? Which removes citric acid?*

Part D: Mixture logic (2 pts)

A sample half-dissolves in water. The dissolved part is neutral and iodine- negative. The undissolved part turns iodine blue-black. Is this one powder or a mixture? Name the component(s).

Part E: Answer format (2 pts)

Rewrite this as a full scoring answer: "*Sample 4 is baking soda.*" (Assume the suspect who bakes is "Suspect B.")

Answer Key

Part A: 1. Calcium carbonate / chalk — $CaCO_3$. 2. Table salt — $NaCl$. 3. Corn starch — $(C_6H_{10}O_5)_n$. 4. Baking soda — $NaHCO_3$. 5. Sucrose (sugar) — $C_{12}H_{22}O_{11}$.

Part B: 1. Steps 1→2 → **chalk** (fizzes, doesn't dissolve). 2. Steps 1→3→4→5 → **sugar**. 3. Steps 1→3 → **cornstarch**.

Part C: One good key:

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1. Turns iodine blue-black?      yes → CORN
STARCH        no → 2
2. Water solution is acidic (pH < 5)?  yes →
CITRIC ACID   no → 3
3. Crystals cubic (vs. needle-like), not bitter?
yes → TABLE SALT   no → EPSOM SALT
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Part D: A mixture — table salt (soluble, neutral, iodine-negative) + corn starch (insoluble, iodine-positive).

Part E: "Sample 4: baking soda (sodium bicarbonate) — $NaHCO_3$ — implicates Suspect B, who bakes."